Lecture 19: N-Variate Joint PDF/PMF/CDF

CSE400 – Fundamentals of Probability in Computing

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1. Joint Distribution of N Random Variables

The joint distribution describes the probability behavior of multiple random variables considered together.

Vector Representation

For random variables $X_1, X_2, \ldots, X_N$, the joint behavior is expressed as:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Representation Methods

Joint distributions can be represented using:

- Joint Probability Mass Function (PMF)
- Joint Probability Density Function (PDF)
- Joint Cumulative Distribution Function (CDF)

2. Jointly Gaussian Random Variables

The joint behavior of variables influenced by many random environmental processes can often be approximated by a Joint Gaussian Distribution.

2.1 Bivariate Gaussian PDF

For two random variables X and Y, the joint PDF is:

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\right.\right.$$

Parameters

- μ_X, μ_Y: Means of X and Y
- σ_X, σ_Y: Standard deviations
- ρ: Pearson correlation coefficient

2.2 Key Properties

Marginal Distributions:

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

Marginal PDF Calculation:

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dx$$

3. Worked Example: Jointly Gaussian RVs

Given:

$$\mu_X = 2,\ \sigma_X^2 = 1,\ \mu_Y = -3,\ \sigma_Y^2 = 4,\ \mathrm{cov}(X,Y) = -1$$

(a) Joint PDF

Standard Deviations:

$$\sigma_X = 1, \quad \sigma_Y = 2$$

Correlation Coefficient:

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{-1}{2} = -0.5$$

Final Joint PDF:

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{3}} \exp\left\{-\frac{2}{3}\left[(x-2)^2 + \frac{(x-2)(y+3)}{2} + \frac{(y+3)^2}{4}\right]\right\}$$

(b) Marginal PDFs

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-2)^2}{2}\right)$$

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(y+3)^2}{8}\right)$$

(c) Probability Pr$(X < 0)$

$$\Pr(X < 0) = F_X(0)$$

$$\Pr(X < 0) = \Pr\left(\frac{X-2}{1} < \frac{0-2}{1}\right) = \Pr(Z < -2) = \Phi(-2)$$

Using symmetry and Q-function:

$$\Phi(-x) = 1 - \Phi(x) = \Pr(Z > x)$$

$$Q(x) = \Pr(Z > x)$$

$$\Pr(X < 0) = Q(2)$$

4. Applications and Context

- Climate Systems: Relationship between atmospheric CO_2 and global temperature
- Wireless Communication: Large MIMO Systems with M transmit and N receive antennas